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AN ANOMALOUS PERMIAN POLE FROM CORSICA

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SUMMARY

An anomalous Permian pole lying at 31.5 °N 124 °W was computed from measurements on a collection of igneous rocks from Corsica. It is suggested that the cause of the anomaly may have been a tectonic rotation associated with the Alpine orogeny.

INTRODUCTION

During the summer of 1961 one of the authors took advantage of the somewhat restricted opportunities which arose during a visit to Corsica to collect oriented samples of the Permian volcanic-sedimentary series exposed on the western side of the island on the Osani peninsula. The measurements of remanence and stability tests were carried out in Newcastle.

Although the total number of specimens is small and their directions of remanence somewhat scattered, the results confirm to the pattern of anomalous Permian poles south of the Alpine front recorded by VAN HILTEN (1962), DE BOER (1963) and other Dutch workers.

GENERAL GEOLOGY

Corsica and Sardinia lie on a submerged platform connected to the coast of Italy and separated from France by a basin exceeding 2,000 m in depth. The general geology of the island was reviewed by DENIZOT (1952) and BROUWER (1957), whilst BODENHAUSEN (1955) made a detailed study of the Osani region. The island can be roughly divided into an eastern and a western zone, separated by a zone of thrusting. The backbone of the western zone is a granitic massif. This massif is unconformably overlain by the volcanic sedimentary series which has at its base a thin coal of Stephanian or possibly late Upper Westphalian age. The volcanic sequence is considered to represent the final stages of the Hercynian orogeny and

consists of a few hundred metres of andesitic lavas and tuffs which culminate in rhyolite. The sequence is generally assigned to the Permian, although clearly a considerable portion may belong to the Stephanian.

Subsequent movements have caused some faulting and imparted a westerly dip (N252°E 35°W) to the volcanics, resulting in a wide area of outcrop on the Osani peninsula.

COLLECTION AND MEASUREMENT OF SAMPLES

Eighteen samples of andesitic lava and ignimbritic tuffs were collected over an area of half a square kilometre and through a vertical height of 200 m on the Osani peninsula. Single samples were collected at each site, and an estimated thickness of 600 m was sampled. A further six samples were also collected from the region east of Manso but, since half of these samples proved too weakly magnetized for measurement on a low sensitivity astatic magnetometer, they were not considered further.

A single 2.5 cm cylinder of 2.5 cm diameter was cut from each sample. The direction and intensity of natural remanent magnetization were recorded using a short period astatic magnetometer (COLLINSON et al., 1957). The cylinders were subjected to progressive demagnetization in alternating fields up to 250 Oe. Three samples proved to be too weak for measurement after the initial demagnet-

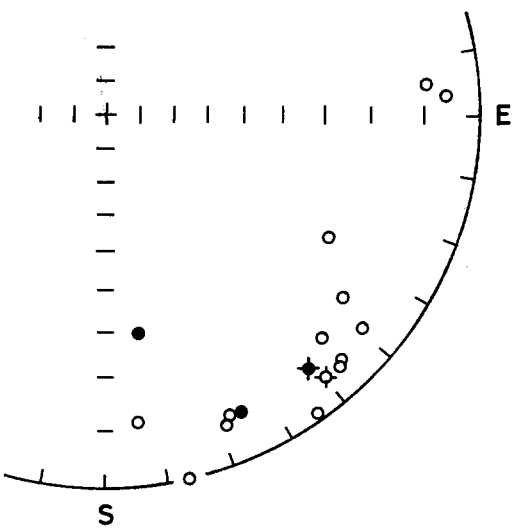


Fig.1. Directions of magnetization corrected to the horizontal and demagnetized in an alternating field of 255 Oe. Closed circles represent downward inclination, open circles upward inclination. Crossed open circle is mean direction with respect to the horizontal, and crossed closed circle is mean direction after correction for geological dip.

TABLE I

SUMMARY OF CORSICAN DATA

Site	Osani peninsula 42.3 °N 8.6 °E
Rocks	Permian or late Carboniferous andesitic lavas and ignimbritic tuffs
Mean declination	141 °
Mean inclination	+ 8.5 °
Circle of confidence	$\alpha = 14°$
Scatter	$k = 8$
Samples (and sites)	used fifteen, three rejected as too weakly magnetized
Ancient pole (S)	31.5 °N 124 °W, $d\psi = 14°$, $d\chi = 7°$
Intensity of magnetization	10^{-4}–10^{-5} e.m.u./cm^3
Susceptibility (two samples)	10^{-5} e.m.u./cm^3

ization stage. The mean direction and scatter for the remainder was computed after each demagnetization stage. The best estimate of the true original direction of magnetization was regarded as that stage at which the scatter was a minimum (after 250 Oe.) The statistic used was FISHER's (1953) dispersion on a sphere which is standard in palaeomagnetism, although the scatter of recorded directions is not strictly Fisherian. The results are illustrated in Fig.1 and are summarized in Table I.

DIVERGENCE OF THE CORSICAN POLE

The mean direction of magnetization, and hence the pole position computed from it, differ significantly from the results from rocks of Permian age collected from sites in northern Europe. Poles calculated from British, German, Norwegian, Czech and Polish rocks show a close-grouping about a mean position of 43 °N 169 °E (NAIRN, 1960; VAN HILTEN, 1962), whereas the Permian pole from Corsica lies at 31.5 °N 236 °E.

There are several possible explanations for the anomaly. It could be argued that the results are an outcome of the method or techniques used. Although the total number of samples is small, there are probably sufficient sites over a sufficient thickness to make reasonable the assumption that secular variation has been averaged out.

The demagnetization procedure, a standard technique, makes it clear that there remains no viscous magnetization, but does not rule out the possibility that the direction may be a resultant of a secondary component as hard magnetically as the primary. A second argument is that should the rocks be strongly anisotropic, then the direction of original magnetization may have been different from

the direction of the geomagnetic field at the time, or that the magnetic vector was rotated out of the original direction by stress induced anisotropy. However, studies of the anisotropy of susceptibility (STONE, 1963) on two samples indicated that differences between the major and minor axes were of the order of 10% of the intermediate axes. It is suggested, therefore, that susceptibility is not a likely cause.

The third, and in the authors' opinion the most likely interpretation, is that the rocks were originally magnetized in a direction which would be consistent with northern European results and that the anomaly arises from post-magnetization movement.

An inspection reveals that all measurements of Permian rocks from regions south of the Alps are anomalous, and VAN HILTEN (1962) and others have attributed this to tectonic rotation and displacement—the latter because rotation about a vertical axis could not account for the dip anomaly. The attraction of the interpretation of the anomalous results in terms of a tectonic effect is that the one geological factor common to all the known areas, i.e., Bolzano, the Estérel, Corsica and the Pyrenees, is their association with Alpine tectonism.

THE TECTONIC ROTATION INTERPRETATION

Whilst any interpretation on the basis of the present sampling must remain tentative, it is interesting to examine the implications of tectonic rotation. Assuming the direction of magnetization to have been originally parallel to the direction predicted from measurements in northern and central Europe, the direction recorded (see Table II) would have required an anticlockwise twist of the Corsican Hercynian block through an angle of 53° ± 14° about a hinge point close to Genoa (see Fig.2) to bring it to its present position.

TABLE II

COMPARISON OF DIRECTIONS OF MAGNETIZATION AND THE MEAN EUROPEAN PERMIAN FIELD[1]

Region	*Magnetization and field*	*Decl.*	*Incl.*	α_{95}	*N*	*Reference*
Corsica	Remanent magnetization	141°	+8.5°	14°	15	NAIRN, 1960
	Supposed Permian field	194°	+6°			VAN HILTEN, 1962
Estérel	Remanent magnetization	210°	−16°	11.5°	5	ROCHE, 1957
		217°	−22.5°	4.5°	14	RUTTEN et al., 1957
		207.5°	−16°	5°	14	As and ZIJDERVELD, 1958
	Supposed Permian field	193°	+4°			NAIRN, 1960
						VAN HILTEN, 1962

[1] The Permian field is based on a mean European Permian pole at 43°N 169.5°E (NAIRN, 1960; VAN HILTEN, 1962).

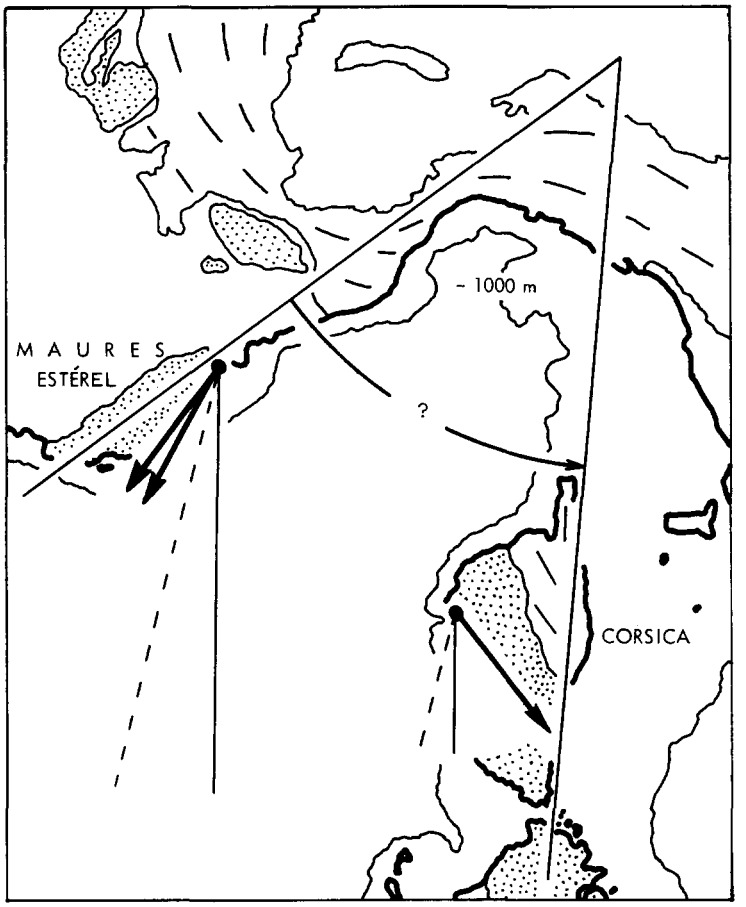


Fig.2. Horizontal components of magnetization of rocks from Corsica and the Maures Estérel. Stippled areas represent Hercynian blocks, and general structural trends are indicated by broken lines. The direction of the horizontal magnetic components are shown by arrows in comparison with the present meridian (full lines) and the Permian meridian (dashed lines) indicated by measurements in northern Europe.

The amount of this rotation, which presumably took place during the Alpine orogeny, is such as to close up the basin now existing between Corsica and the French coast. This is one interpretation consistent with the palaeomagnetic measurements and it is therefore of some interest to note that CAREY (1958) considered the basin opened up during the Alpine orogeny from a consideration of orogenic belt trends. The similarity of the two results suggests that detailed investigation, both of tectonics and palaeomagnetism, ought to give a clear indication of the feasibility of the proposed rotation.

The Maures Estérel massif, which in this interpretation would oppose the Hercynian block of Corsica, appears only to require a slight rotation in the opposite sense (ROCHE, 1957; RUTTEN et al., 1957; AS and ZIJDERVELD, 1958). It will be

seen from Table II that, in contrast to the Corsican result, the discrepancy between the angle of inclination reported and the predicted value is well in excess of the confidence limits of the mean inclination. On the basis of similar inclination discrepancies Dutch workers, such as VAN HILTEN (1962), who regard them as real effects, have suggested large translational movements associated with rotation in other areas, particularly in northern Italy. Simple rotation about an inclined axis would seem to imply impossibly large vertical movements.

CONCLUSIONS

There is an interesting apparent agreement between the proposed interpretation of the palaeomagnetic measurements, the rotation of Corsica, and CAREY's (1958) geotectonic arguments which ought to be amenable to further investigation. If it is true that tectonic rotation of this magnitude can occur, then no interpretation can be made without consideration of the tectonic framework in which the rocks occur. This limitation is not normally given too much consideration in discussion of the criteria of suitability of rocks for palaeomagnetic research—lithology and unmetamorphosed condition, etc. Although, as in this case, there is plenty of evidence of disturbance indicated by minor folding and faulting, there may be little sign of large scale rotation.

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